Same-field pivot transport

2026-11-01 · Derived identity and syntax-checked source; zero numerical trials

Exact construction. Given $Z_a A - A Z_b = G K^t$, let $u=e_i-e_j$ and $P=I-u u^t$ be a row transposition. Then $B=PA$ has generators

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GB = [P G | u | -Z_a u]
KB = [K   | A^t Z_a^t u | A^t u]
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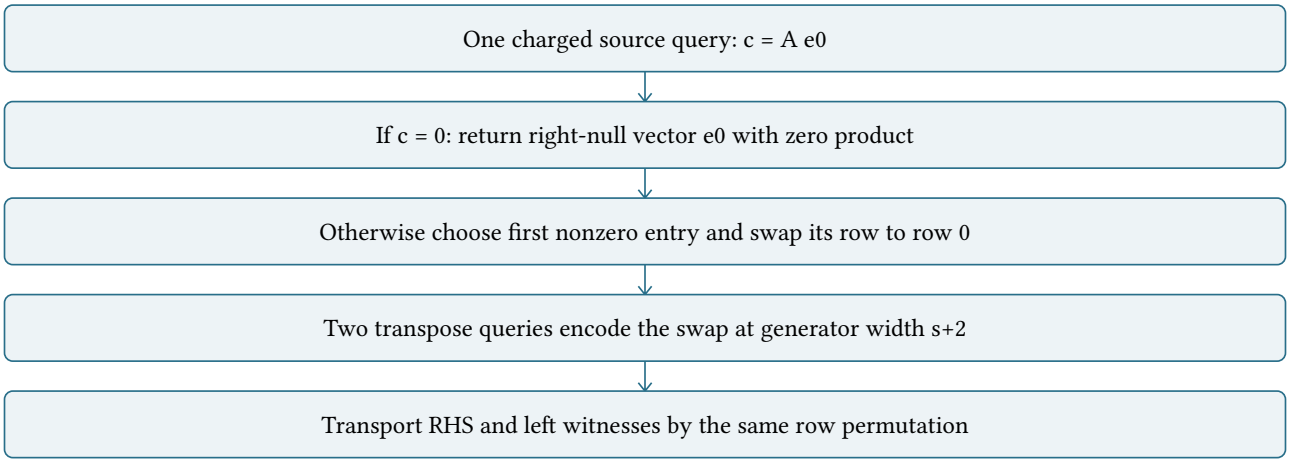
The identity follows from $Z_a P-P Z_a = u u^t Z_a - Z_a u u^t$. Constructed width grows by two. Two charged transpose source queries obtain the added right columns. The base prime and wraps stay unchanged. An identity swap makes no source queries. Outside the query ledgers, the implementation charges two field products and N+1 subtractions.



Pivot guarantee at one position. Every nonsingular A has a nonzero first column. Its first nonzero row supplies a nonzero scalar pivot after a swap, with no randomized field-size condition. Total queries: one when the top entry is already nonzero; three for a nontrivial swap. An all-zero first column yields the actual right-null vector e0. It does not establish RHS inconsistency or produce a left-null exclusion witness. Scan length is an explicit native counter. The diagram's zero-column case terminates; only the nonzero branch proceeds to row swapping.

Certificate transport. $Ax=b$ is equivalent to $(PA)x=P b$. Solution and right-null coordinates stay unchanged. A left-null vector w for PA maps to $P^t w$ for A, preserving $(P^t w)^t b = w^t(Pb)$. Feature-nullness needs its own check. The invertible swap matrix with zero leading scalar is a required control: this pivot maps it to the identity.

Cost dependency. One scalar pivot does not repair an arbitrary balanced leading block. Repeated row pivots plus scalar Schur reduction can grow constructed width to $s+4d$ after d levels. This helper is not inserted as an uncharged fallback into the balanced recursive inverse. Rank-revealing block selection or controlled generator compression is the next construction. Audit histories, copies and temporary encodings count towards memory.

Validation next. Freeze all declared transpositions, both query directions, zero and later pivots, boundary rows, wrap guards, corrupted correction signs, RHS and witness transport, and exact query/scalar/scan counters. No trials precede the full upstream validation gates and a committed protocol.

Files and prior art. pivot.py and DERIVATION.md. Bostan et al. (2017), Section 6, DOI 10.1137/16M1062855, motivates explicit rank-profile handling; its randomized regularization guarantee is not assumed here. This is a derived pivot primitive, not a measured speed result.